CATERPILLAR®

DEALER BEST PRACTICE SERIES

Reduction of Internal Contamination for Hoses

Maintenance and Repair | Application | **Component Life Management** | Component Renewal (CRC) | MARC Management

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Although Caterpillar does not endorse this particular supplier, Marcosa has experience with the following products or companies. We welcome additional supplier recommendations from our dealer network.

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1.0 Introduction

The purpose of this Best Practice (BP) is to describe a new hose sealing system through a high efficient biodegradable plastic seal to reduce internal contamination during storage and transportation. **It should be noted that Caterpillar's approved plastic caps should be used in tandem with this BP. Currently, Caterpillar is in the process of designing better plastic caps at which time this BP will be void.**

2.0 Best Practice Description

The cylindrical plastic seals are placed at the ends of the hose and fixed through a heating process using thermal retractable equipment such as a heat gun. A complete seal is attained in less than 4 seconds, and the removal of the plastic seal occurs easily within a few seconds. The use of this procedure ensures that there will be very minimal internal contamination.

3.0 Implementation Steps

1. Clean both ends of the hose before applying the plastic plug.
2. Apply Caterpillar's approved plastic caps to ensure protection. The new plastic wrap will shrink OVER the Caterpillar approved plastic cap.
3. Wear appropriate PPE as this process will expose you to high temperatures.
4. It is up to each individual dealership to create the proper mounting device for the heating equipment.
5. Turn on the thermal retractable equipment and let it heat for roughly one minute to obtain the ideal working temperature; heating times will vary depending on type of heating equipment. For portable equipment such as this heat gun, it is recommended to work between positions 3 and 4 (average temperature of 400ºC/752ºF).
6. Choose the appropriate plastic cylindrical plug to slide over the end of the exposed hose, Figure 1.
7. Slide the plug over the end of the hose and use the heat gun to seal, Figures 2 and 3.
8. Figure 4 and 5 show the hose end sealed with plastic.
9. To remove the seal, Figure 6, simply hold the tip of the black strip, which is part of the original plastic cylindrical seal as seen in Figure 1, and pull.

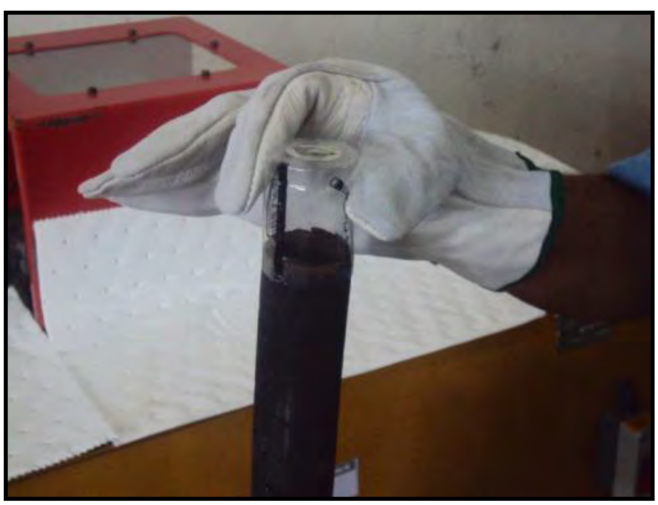

Figure 1

Figure 2

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	DATE	CHG NO	NUMBER
Reduction of Internal Contamination for Hoses	2/17/2015	00	1111-3.03-1263

Figure 3

Figure 4

Figure 5

Figure 6

Marcosa Cat has a 5 star ranking in contamination control by Caterpillar's assessment.

4.0 Benefits

- Reduced time spent on the sealing process.
- Reduced time removing the seals.
- Protection against internal contamination.
- Reduced cost.
- Recyclable seals.
- Ensures that the seals will not come off before the final destination.

5.0 Resources Required

Plastic Plugs:

Thermal retractable equipment (Heat Gun) + Diffuser.

6.0 Supporting Attachments / References

- Cost estimates for seals are listed below. Converstion to USD is available.

Seal	Sizes		Amount	Unit Value	Total Value $	Total Value R$
seal exclusive	25 x 40	Ø 12-2	1000	$ 0,18	$ 182,91	R$ 370,00
seal exclusive	28 x 40	Ø 22-2	1008	$ 0,19	$ 194,33	R$ 393,12
seal exclusive	34 x 40	Ø 27-3	560	$ 0,20	$ 113,50	R$ 229,60
seal exclusive	38 x 50	Ø 30-3	550	$ 0,21	$ 114,19	R$ 231,00
seal exclusive	46 x 60	Ø 32-4	516	$ 0,21	$ 109,68	R$ 221,88
seal exclusive	80 x 60	Ø 60-7	434	$ 0,53	$ 229,56	R$ 464,38
					$ 944,18	R$ 1.909,98

1$ = R$ 2,0229
05/07/2012

- Video demonstration by Ultra Clean
  - http://www.youtube.com/watch?v=jKYWW421uX4&feature=player_embedded

- Instruction Manual - UltraClean EN.pdf
- Supplier for materials is Ultra Clean
  - Ultra Clean Systems
  - www.ultracleanbrasil.com.br
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7.0 Related Best Practices

None

8.0 Acknowledgements

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